Lecture 4. Least Squares: QR Decomposition

Gram–Schmidt, Householder, Givens, conditioning of least squares, underdetermined problems

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4.1 QR decomposition and least squares

Cholesky on the normal equations is one way to do least squares; the QR decomposition is another, and is what R's `lm` uses. Let $X \in \mathbb{R}^{n\times p}$ have full column rank ($n \geq p$).

Definition (QR decomposition)

- **Full QR**: $X = QR$ with $Q \in \mathbb{R}^{n\times n}$ orthogonal and $R \in \mathbb{R}^{n\times p}$ upper triangular (its lower $(n-p)\times p$ block is 0). The first p columns of Q are an orthonormal basis of the column space $\mathcal{R}(X)$, the last $n-p$ of $\mathcal{N}(X^\top) = \mathcal{R}(X)^\perp$.
- **Reduced QR**: $X = Q_1 R_1$ with $Q_1 \in \mathbb{R}^{n\times p}$, $Q_1^\top Q_1 = I_p$ (but $Q_1 Q_1^\top \neq I_n$), and $R_1 \in \mathbb{R}^{p\times p}$ upper triangular with positive diagonal.

Since $X^\top X = R^\top Q^\top Q R = R_1^\top R_1$, a QR decomposition of X gives the Cholesky factor of the Gram matrix for free.

Least squares via QR. The normal equations $X^\top X\beta = X^\top y$ become $R_1^\top R_1\beta = R_1^\top Q_1^\top y$, and since R_1 is invertible,

$$R_1\beta = Q_1^\top y,$$

a single back substitution. (Equivalently, $\|y - X\beta\|^2 = \|Q^\top y - R\beta\|^2 = \|Q_1^\top y - R_1\beta\|^2 + \|Q_2^\top y\|^2$, minimized when the first term vanishes, and the residual sum of squares is $\|Q_2^\top y\|^2$.) This is more accurate than solving the normal equations because

$$\kappa(X^\top X) = \kappa(X)^2,$$

since the singular values of $X^\top X$ are the squares of those of X. Standard errors need $(X^\top X)^{-1} = R_1^{-1}R_1^{-\top}$, again via triangular solves.

4.2 Gram–Schmidt

Every full-column-rank X has a reduced QR decomposition: it is the Gram–Schmidt orthonormalization of the columns $x_1, \ldots, x_p$.

(1) $q_1 = x_1/\|x_1\|_2$.
(2) For $k = 2, \ldots, p$: $v_k = x_k - \sum_{j<k}\langle q_j, x_k\rangle q_j$ and $q_k = v_k/\|v_k\|_2$.

The q_k are orthonormal with $\operatorname{span}\{x_1, \ldots, x_k\} = \operatorname{span}\{q_1, \ldots, q_k\}$ for each k. With $r_{jk} = \langle q_j, x_k\rangle$ $(j < k)$, $r_{kk} = \|v_k\|_2$ and $r_{jk} = 0$ $(j > k)$, step (2) reads $x_k = \sum_{j\leq k} r_{jk} q_j$, i.e. $X = Q_1 R_1$.

Classical vs modified. Classical Gram–Schmidt (CGS) computes all $\langle q_j, x_k\rangle$ with the original x_k. When columns are nearly collinear, rounding errors make the computed q's far from orthogonal. **Modified** Gram–Schmidt (MGS) subtracts the projections one at a time, projecting the *current* partially orthogonalized vector:

$$v_k^{(1)} = x_k, \qquad v_k^{(j+1)} = v_k^{(j)} - \langle q_j, v_k^{(j)}\rangle q_j, \quad j = 1, \ldots, k-1, \qquad q_k = v_k^{(k)}/\|v_k^{(k)}\|.$$

In exact arithmetic the two agree; in floating point MGS loses much less orthogonality, though it is not fully immune. Both cost $\sum_k 4n(k-1) \approx 2np^2$ flops and give only the reduced QR.

4.3 Householder reflections

Proposition 4.1 (Householder reflector)

For $x, w \in \mathbb{R}^n$ with $\|x\|_2 = \|w\|_2$ and $x \neq w$, let $u = (x-w)/\|x-w\|_2$ and $H = I - 2uu^\top$. Then H is symmetric and orthogonal and $Hx = w$.

Proof. $H^\top = H$ and $H^2 = I - 4uu^\top + 4u(u^\top u)u^\top = I$. Next, $\|x-w\|^2 = 2x^\top x - 2w^\top x$ (using $x^\top x = w^\top w$), so

$$Hx = x - 2\frac{(x-w)^\top x}{\|x-w\|^2}(x-w) = x - \frac{x^\top x - w^\top x}{x^\top x - w^\top x}(x-w) = w.$$

■

Geometrically H reflects across the hyperplane orthogonal to u. Householder QR chooses H_1 with $w = \pm\|x_1\|e_1$ to zero the first column below the diagonal, then $H_2 = \mathrm{diag}(1, \tilde{H}_2)$ acting on rows 2:n to zero the second column, and so on:

$$H_p \cdots H_2 H_1 X = \begin{pmatrix} R_1 \\ 0 \end{pmatrix}, \qquad Q = H_1 H_2 \cdots H_p.$$

This gives the *full* QR. The sign of w is chosen opposite to the first entry of x, to avoid cancellation in $x - w$.

Implementation and cost. The matrices H_j are never formed: $HX = X - 2u(u^\top X)$ costs $4np$ flops. At stage j, computing $\tilde{u}_j$ costs $3(n-j+1)$ and applying $\tilde{H}_j$ to the $(n-j+1)\times(p-j+1)$ block costs $4(n-j+1)(p-j+1)$; in total

$$\sum_{j=1}^{p} 4(n-j+1)(p-j+1) \approx 2np^2 - \tfrac{2}{3}p^3 \text{ flops.}$$

The algorithm works in place: R_1 overwrites the upper triangle, and the Householder vectors (normalized so their first entry is 1) are stored below the diagonal. To compute $Q^\top y$ or Qv one applies the reflections successively; forming Q explicitly costs another $2np^2 - \frac{2}{3}p^3$. Householder QR is backward stable and is what LAPACK (`geqrf`, or `geqp3` with column pivoting) and R's `qr` use.

4.4 Givens rotations

A Givens rotation $G(i,k,\theta)$ is the identity except for the entries $c, s, -s, c$ in rows and columns i, k, with $c = \cos\theta$, $s = \sin\theta$; it is orthogonal. For $y = G^\top x$,

$$y_i = cx_i - sx_k, \quad y_k = sx_i + cx_k, \quad y_j = x_j \ (j \neq i,k),$$

and the choice $c = x_i/\sqrt{x_i^2 + x_k^2}$, $s = -x_k/\sqrt{x_i^2 + x_k^2}$ makes $y_k = 0$. Applying $G^\top$ to a matrix changes only two rows ($6m$ flops). QR by Givens zeros one entry at a time, $G_t^\top \cdots G_1^\top X = \binom{R_1}{0}$, at a cost of about $3np^2 - p^3$ flops: more than Householder, but well suited to sparse or structured matrices (e.g. updating a QR when a row is added), since it touches only the entries it needs. Each rotation can be stored as a single number in the zeroed position.

4.5 Conditioning of least squares

For a nonsquare X, $\kappa(X) = \sigma_{\max}(X)/\sigma_{\min}(X) = \sigma_{\max}(X)\sigma_{\max}(X^\dagger)$, with $X^\dagger$ the Moore–Penrose pseudoinverse. The condition number of the least squares *problem* depends on the residual; it is

$$\kappa(X) + \frac{\kappa(X)^2 \tan\theta}{\eta}, \qquad \cos\theta = \frac{\|X\beta\|}{\|y\|}, \quad \eta = \frac{\|X\|\|\beta\|}{\|X\beta\|}.$$

When X is nearly collinear and the residuals are not small, the $\kappa(X)^2$ term dominates, so stable algorithms matter.

Example 4.1

$X = \begin{pmatrix} 1 & 1 \\ 10^{-3} & 0 \\ 0 & 10^{-3} \end{pmatrix}$ has full rank, but $X^\top X = \begin{pmatrix} 1+10^{-6} & 1 \\ 1 & 1+10^{-6} \end{pmatrix}$ rounds to the singular $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ with six significant digits. Forming the normal equations destroys the information, while QR applied to X itself does not.

method	flops	stability	notes
Cholesky on $X^\top X$	$np^2 + \frac{1}{3}p^3$	least stable	fastest; $X^\top X$ can be updated with streaming data
Householder QR	$2np^2 - \frac{2}{3}p^3$	stable	default in R, MATLAB, Julia
MGS	$2np^2$	stable-ish	gives Q_1 explicitly
SVD	$4n^2p + 8np^2 + 9p^3$	most stable	also handles rank deficiency

There is no universal gold standard: Cholesky is twice as fast for $n \gg p$ and needs only $X^\top X$ (which fits in memory even when X does not), while QR is more accurate.

4.6 Underdetermined least squares

If $\operatorname{rank} X < p$ (in particular $n < p$), the normal equations have infinitely many solutions, and an extra criterion is needed.

- **Minimum ℓ_2 norm**: minimize $\|\beta\|_2$ subject to $X^\top X\beta = X^\top y$. The solution is $\hat\beta = X^\dagger y$, which for full row rank X ($n < p$) is $X^\top(XX^\top)^{-1}y$. It is also the limit of ridge regression: $\hat\beta = \lim_{\lambda\downarrow 0}(X^\top X + \lambda I)^{-1}X^\top y$.
- **Minimum ℓ_0 "norm"**: minimize $\|\beta\|_0 = \#\{j : \beta_j \neq 0\}$, the sparsest solution. This is best subset selection, a combinatorial problem.
- **Minimum ℓ_1 norm**: a convex relaxation (basis pursuit, related to the lasso), solvable by linear programming, which under suitable conditions recovers the sparsest solution.